Machine Learning and Big Data - DATA622

CUNY School of Professional Studies

Big Data

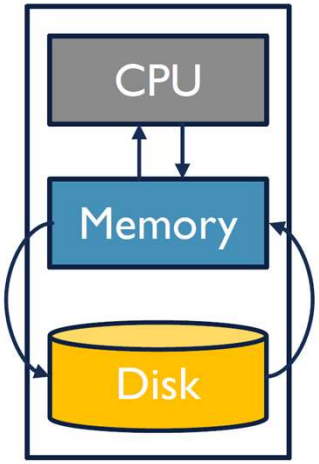
CUNY School of Professional Studies

Big Data

- ***Big data*** is the term for a collection of data sets so large and complex that it becomes difficult to process using on-hand database management tools or traditional data processing applications.
- The challenges include capture, curation, storage, search, sharing, transfer, analysis, and visualization.
- The trend to larger data sets is due to the additional information derivable from analysis of a single large set of related data, as compared to separate smaller sets with the same total amount of data, allowing correlations to be found to "spot business trends, determine quality of research, prevent diseases, link legal citations, combat crime, and determine real-time roadway traffic conditions."

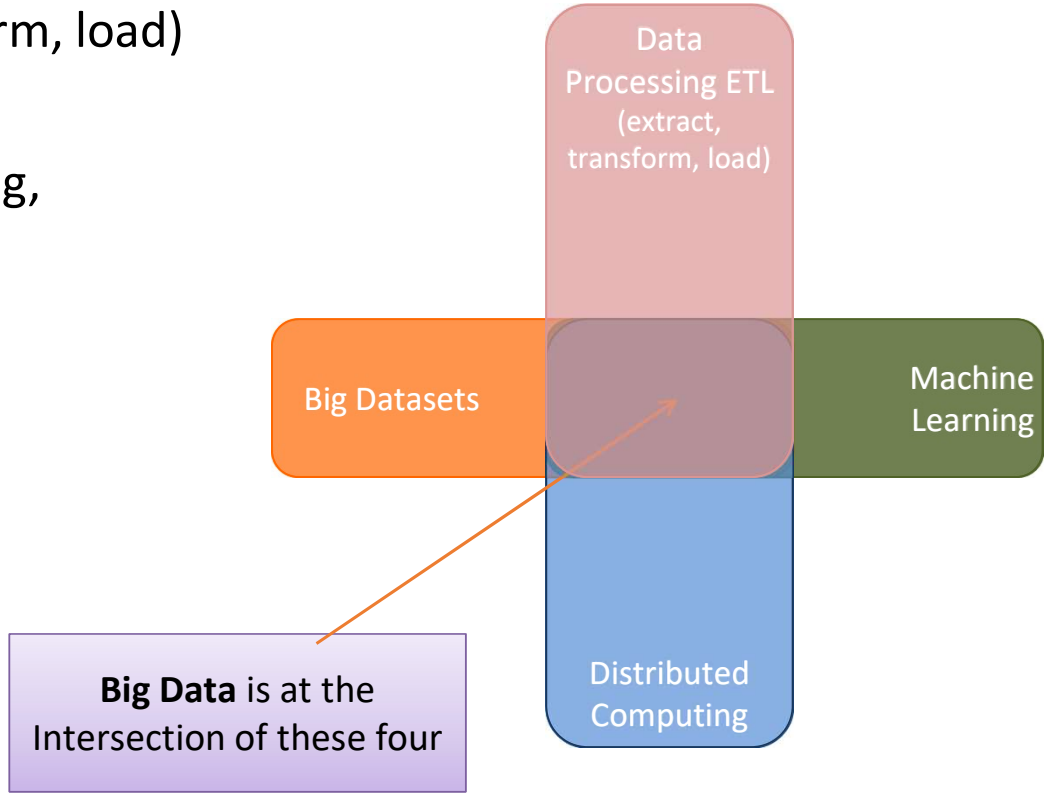
Traditional Data Processing

- Reading a word from disk vs. main memory: 105 slower.
- Normal disks only reach 150MB/s for sequential reads.
- I/O Bounded: biggest performance bottleneck is reading / writing large files to disk.
  - 100 GBs ~ 10 minutes.
  - 200 TBs ~ 20,000 minutes = 13days.



Big Data

- Data processing (third trend)
  - Traditional ETL (extract, transform, load)
  - Data Stores (HBase, ……..)
  - Tools for processing of streaming, multimedia & batch data
- Machine Learning (fourth trend)
  - Classification
  - Regression
  - Clustering
  - Collaborative filtering



Source: Michelle Carney

5-Vs of Big Data

Volume

Data Size & Growth Rate

Variety

Data types

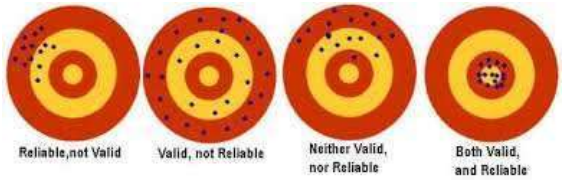
Veracity

Can the elicited results be believed?

Velocity

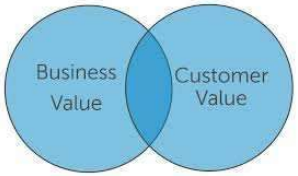
Speed requirements

Validity



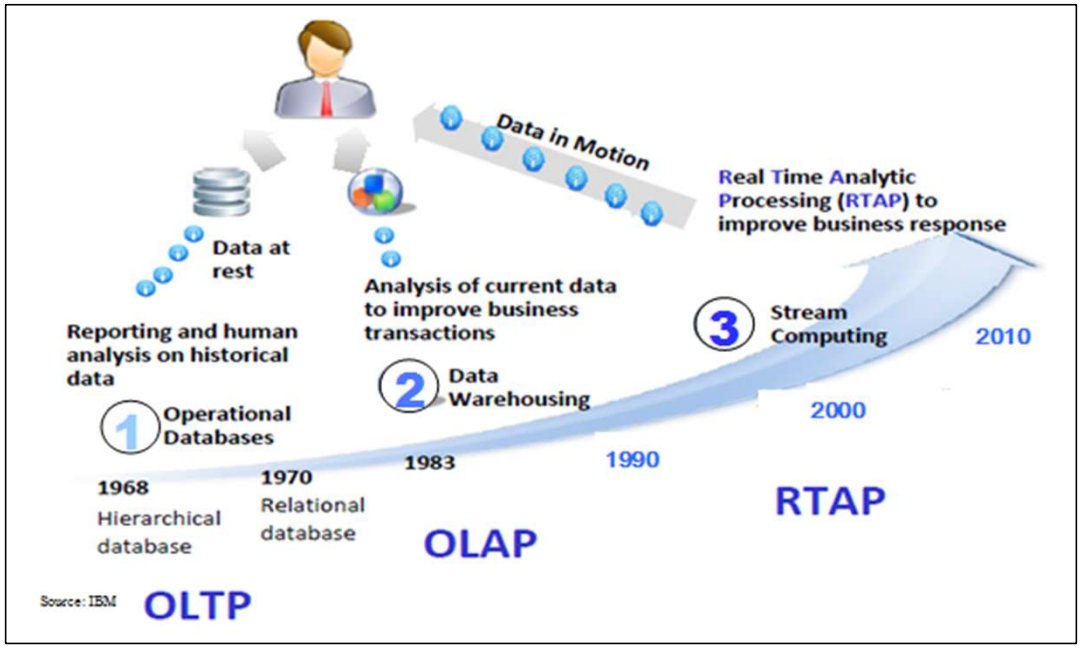
Legitimacy of the data sets (governance provisions)?

Value



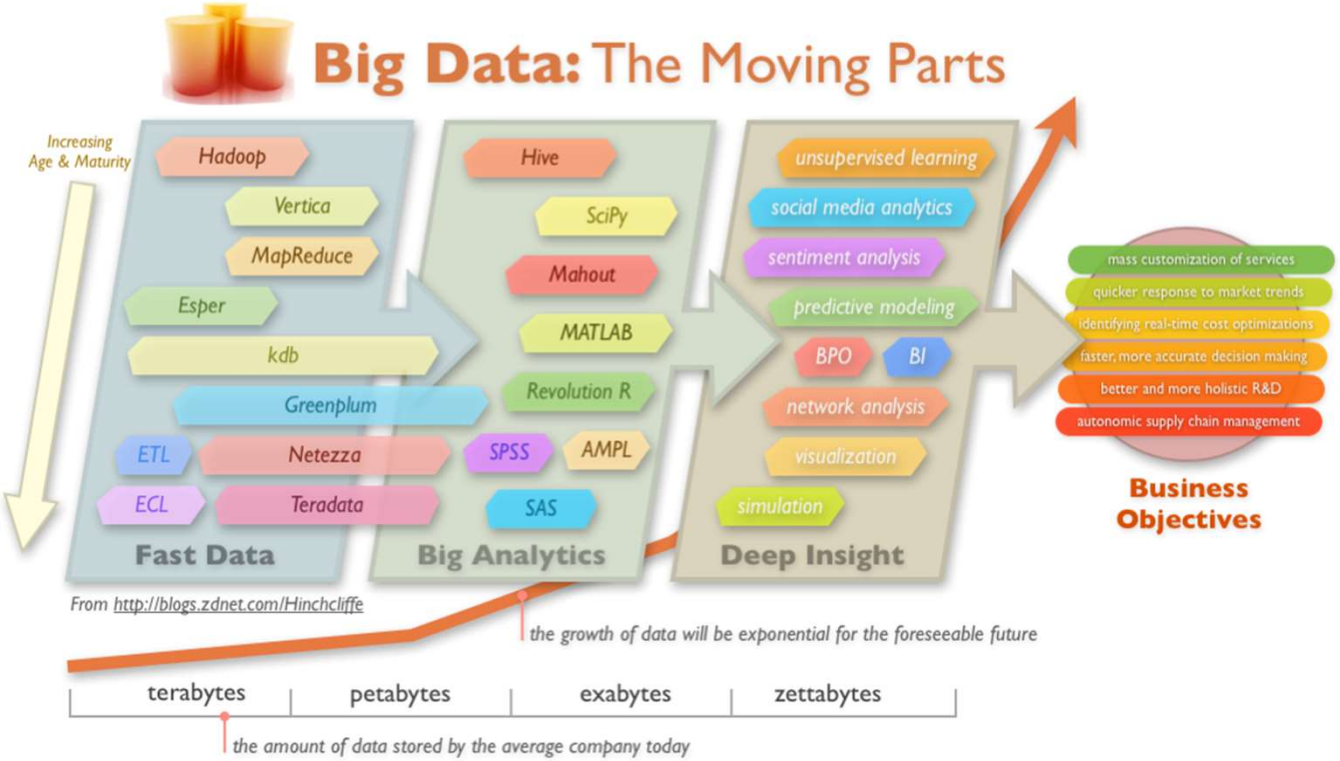
What business advantages can be gleaned?

Harnessing Big Data



- **OLTP:** Online Transaction Processing (DBMSs)
- **OLAP:** Online Analytical Processing (Data Warehousing)
- **RTAP:** Real-Time Analytics Processing (Big Data Architecture & technology)

Big Data Technology



How do we use very large amounts of data?

- Working with big data is *not* about
  - code optimization
  - learning details of todays hardware/software:
    - GraphLab, Hadoop, parallel hardware, ….
- Working with big data *is* about
  - Understanding the cost of what you want to do
  - Understanding what the tools that available offer
  - Understanding how much can be accomplished with linear or nearly-linear operations (e.g., sorting, …)
  - Understanding how to organize your computations so that they effectively use whatever's fast
  - Understanding how to test/debug/verify with large data

Source: Michelle Carney

Source: Michelle Carney

Hadoop

CUNY School of Professional Studies

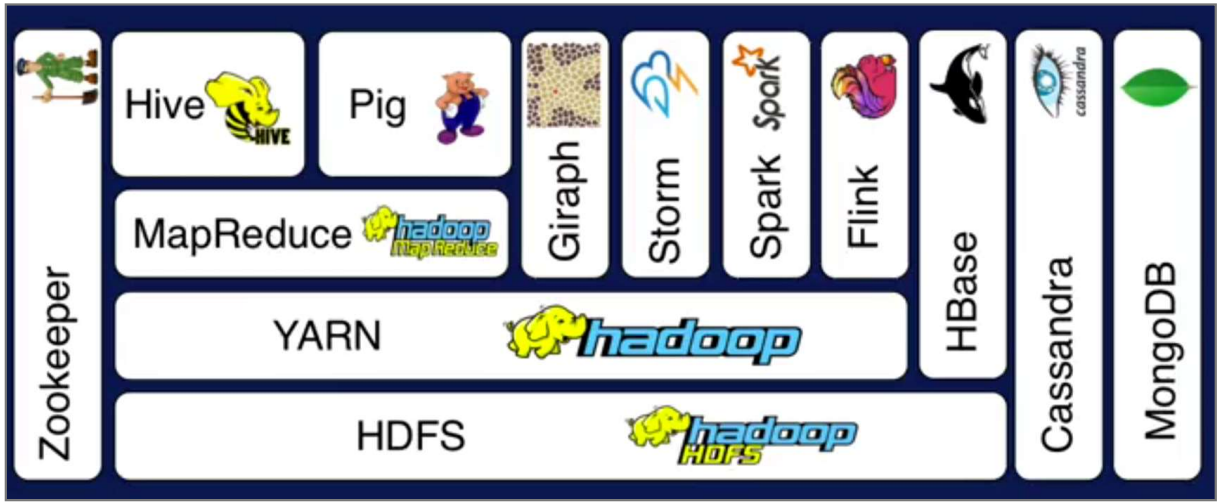
Hadoop

- Hadoop is a framework for storing data on large clusters of ***commodity*** hardware and running applications against that data.
  - No need for big and expensive servers.
  - Many affordable and easily available computers with single-CPU are tied together.
- A ***cluster*** is a group of interconnected computers (we call computers in the cluster as nodes) that can work together on the same problem.
- Using networks of affordable compute resources to acquire business insight is the key value proposition of Hadoop.

Hadoop

CUNY School of Professional Studies

Hadoop Ecosystem

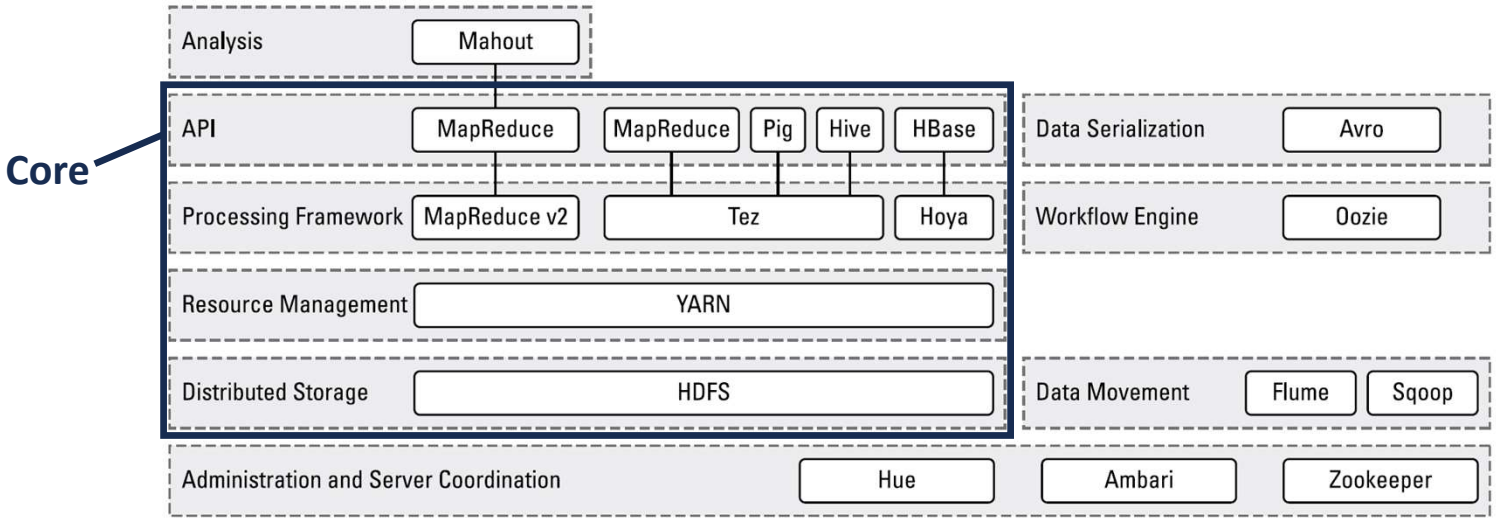


Source: Michelle Carney

Hadoop

- ***Independent Nodes***: Hadoop is a cluster with independent machines (cluster with nodes), that every node perform its job by using its own resources.
- ***Fault tolerance***: When data is sent to an individual node, that data is also replicated (usually x3) to other nodes in the cluster. In the event of failure, there is another copy available for use.
- ***Commodity hardware***: Hadoop does not require a very high-end server with large memory and processing power. Hadoop runs on JBOD (just bunch of disk), so every node is independent in Hadoop.
- ***Horizontal Scaling***: We do not need to improve the power of nodes. As the data keeps on growing, we just keep adding nodes..

Hadoop Ecosystem

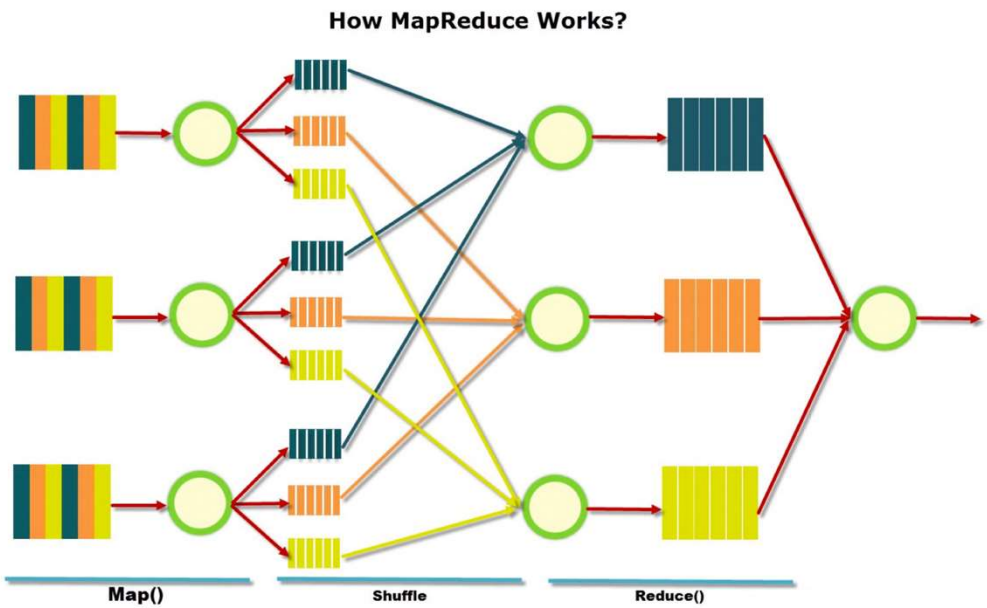


Project Name	*Description*
Hive	A distributed data warehouse for data that is stored in HDFS; also provides a query language that's based on SQL (HiveQL)
Hue	A Hadoop administration interface with handy GUI tools for browsing files, issuing Hive and Pig queries, and developing Oozie workflows
Mahout	A library of machine learning statistical algorithms that were implemented in MapReduce and can run natively on Hadoop
Oozie	A workflow management tool that can handle the scheduling and chaining together of Hadoop applications
Pig	A platform for the analysis of very large data sets that runs on HDFS and with an infrastructure layer consisting of a compiler that produces sequences of MapReduce programs and a language layer consisting of the query language named Pig Latin
Sqoop	A tool for efficiently moving large amounts of data between relational databases and HDFS
ZooKeeper	A simple interface to the centralized coordination of services (such as naming, configuration, and synchronization) used by distributed applications

Source: Michelle Carney

Map Reduce

- MapReduce is a framework for parallel computing.
- It is usually composed of three operations:
  - ***Map***: each node applies the map function to the local data, and writes the output to a temporary storage.
  - ***Shuffle***: nodes redistribute data based on the output keys (produced by the map function), such that all data belonging to one key is located on the same node.
  - ***Reduce***: nodes now process each group of output data, per key, in parallel.

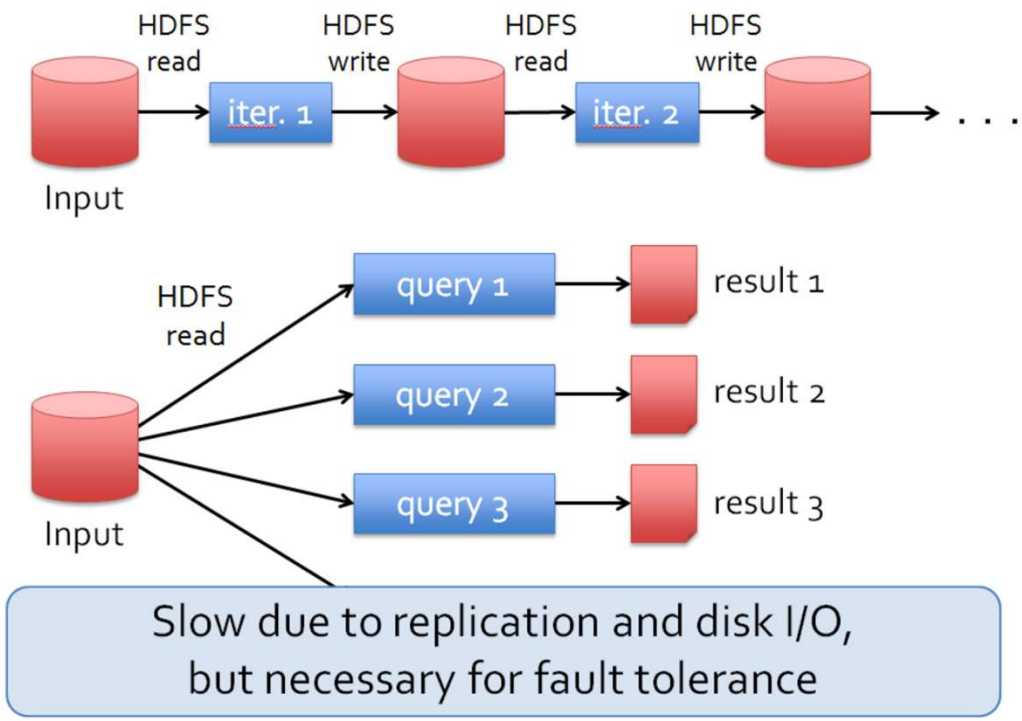


CUNY School of Professional Studies

Limitations of Hadoop

- Lack of long-lived MapReduce jobs
  - MapReduce jobs are typically short-lived
  - One would have to create fresh MapReduce jobs for every iteration in a lot of these classes of computations
- Inability to store a working set of data in memory.
  - The results of every iteration would get stored in HDFS.
    - HDFS -> Map -> Local FS -> Network -> Reduce -> HDFS -> Map ->…
  - The next iteration would need data to be initialized, or read, from HDFS to memory.

Hadoop Ecosystem

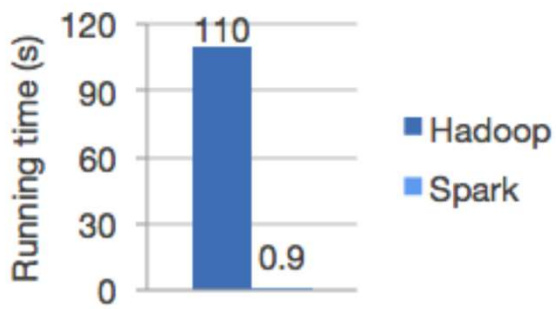


Copied from Matei Zaharia, et.al., Resilient Distributed Datasets, A Fault-Tolerant Abstraction for In-Memory Cluster Computing, amplab, UC Berkeley

Spark

Spark

- Apache Spark is an open source cluster computing framework for real-time data processing.
- The main feature of Apache Spark is its ***in-memory cluster computing*** that increases the processing speed of an application..
- Spark addresses this issue via Resilient Distributed Datasets (RDDs)
- RDD allows applications to keep working sets in memory for reuse.



Logistic regression in Hadoop and Spark

CUNY School of Professional Studies

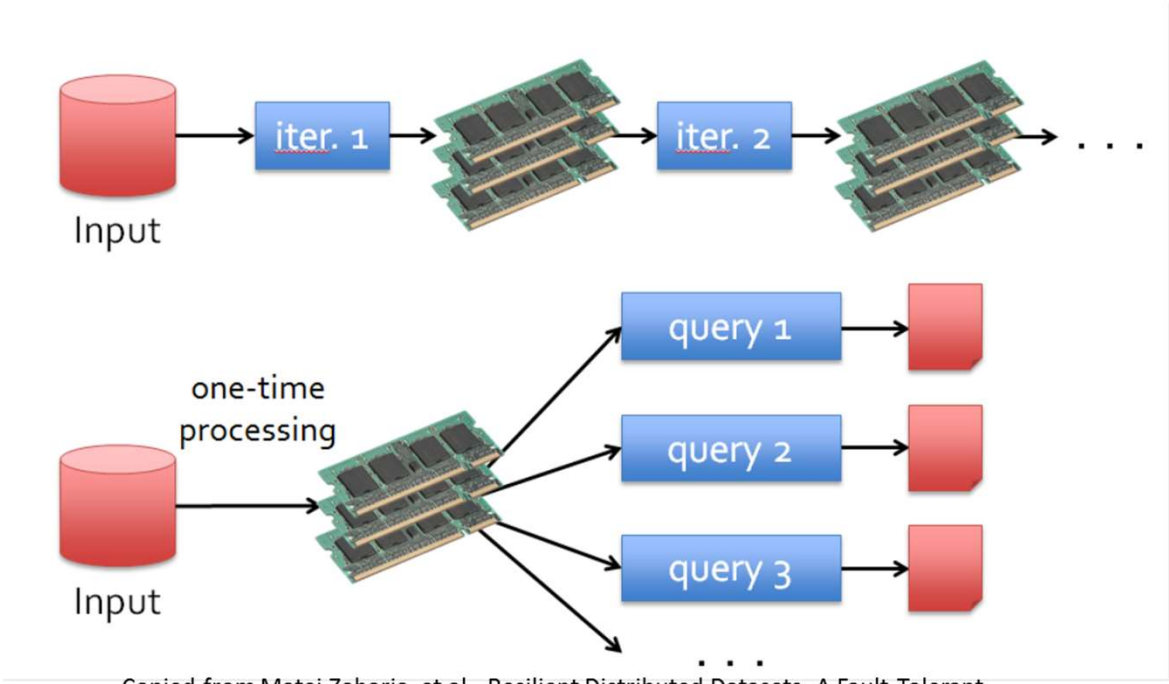
Features of Spark

- ***Polyglot***
  - Provide high-level APIs in Java, Scala, Python and R. Spark code can be written in any of these four languages.
  - For example, Python with PySpark mayb eused for training/development. Scala is usually adopted for industrial big data development.
- ***Speed***
  - Run up to 100 times faster than Hadoop MapReduce for large-scale data processing.
  - Able to achieve this speed through controlled partitioning.
- ***Multiple Formats***
  - Support multiple data sources such as Parquet, JSON, Hive and Cassandra apart from the usual formats such as text files, CSV and RDBMS tables.

Features of Spark

- ***Lazy Evaluation***
  - Delay its evaluation till it is absolutely necessary. This is one of the key factors contributing to its speed.
- ***Real Time Computation***
  - Computation is real-time and has low latency because of its in-memory computation.
- ***Machine Learning***
  - Spark's MLlib is the machine learning component which is handy when it comes to big data processing.
  - It eradicates the need to use multiple tools, one for processing and one for machine learning.
  - MLlib is a distributed machine learning framework on top of Spark core
  - Provides multiple types of machine learning algorithms
    - Classification
    - Regression
    - Clustering
    - Collaborative filtering

Spark Processing



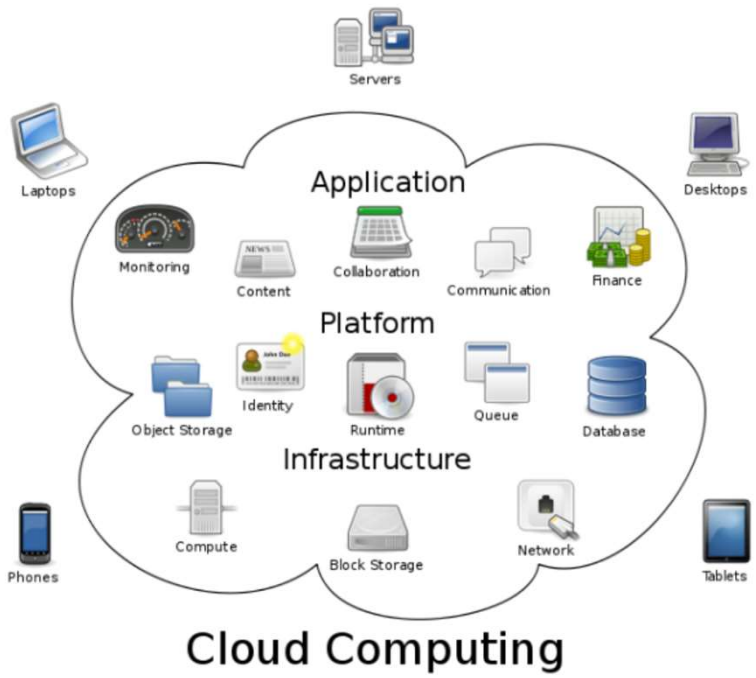
Copied from Matei Zaharia, et.al., Resilient Distributed Datasets, A Fault-Tolerant Abstraction for In-Memory Cluster Computing, amplab, UC Berkeley

But memory is not reliable, how about FT?

Cloud Computing

CUNY School of Professional Studies

Cloud Computing



Cloud Computing

- IT resources provided as a service
  - Compute, storage, databases, queues
- Clouds leverage economies of scale of commodity hardware
  - Cheap storage, high bandwidth networks & multicore processors
  - Geographically distributed data centers
- Offerings from Microsoft, Amazon, Google, …

Benefits

- Cost & management
  - Economies of scale, “out-sourced” resource management
- Reduced Time to deployment
  - Ease of assembly, works “out of the box”
- Scaling
  - On demand provisioning, co-locate data and compute
- Reliability
  - Massive, redundant, shared resources
- Sustainability
  - Hardware not owned